CHAPTER 1

The Liquid Drop Model

1.1 Introduction

The liquid drop model (LDM) of the nucleus was historically the first model to be proposed as an explanation of the different properties of the nucleus. Since it has regained interest in recent times, we begin with a short outline of this phenomenological model. Some aspects of this model will be taken up in the course of the book and explained from a more microscopic point of view. In this chapter we follow to a large extent the standard representation as given in the texts of, for example, J. A. Eisenberg and W. Greiner [EG 70], A. de Shalit and H. Feshbach [SF 74], and A. Bohr and B. R. Mottelson [BM 75].

The idea of considering the nucleus as a liquid drop originally came from considerations about its saturation properties (see below) and from the fact that the nucleus has a very low compressibility and a well defined surface. However, as we shall see, it is misleading to take this point of view too seriously, since in other respects a nucleus does not bear very much resemblance to an ordinary liquid. For instance, the mean distance of two particles in a liquid is roughly given by the value at which the interparticle force has its minimum value, which for nuclei would be $\sim$0.7 fm. However, nucleons in nuclei are, on the average, $\sim$2.4 fm apart. One reason for this big difference as compared with an ordinary liquid is that the nucleons obey Fermi statistics and a nucleus is thus a quantum fluid. The Pauli principle prevents the nucleons from coming too close to one another. Thus scattering events are very scarce in a quantum fluid, whereas in an ordinary fluid they are predominant.

Consequently, the mean free path of the nucleons inside a ground-state (or moderately excited state) nucleus is of the order of the nuclear dimension and resembles, therefore, apart from the statistics, a non-interacting gas. This fact has some drastic consequences that are absent in droplets of ordinary fluids. For instance, we want to mention that in a vibrating nuclear drop [e.g., (dynamical) quadrupole deformations] the momentum distribution of the nucleons may be influenced by the deformations; i.e., it can be anisotropic, whereas the velocity distribution in an ordinary fluid drop is always isotropic. These peculiarities of a quantum drop will be treated in more detail in Chapter 13. Here, instead, we shall follow the historical development of the LDM idea, since it is very useful in describing overall properties of the nucleus and in introducing many concepts of collective phenomena in nuclear physics in a simple way. This is the main purpose of the first chapter, where we will not bother about the specific differences between an ordinary liquid drop and a quantum drop.

1.2 The Semi-empirical Mass Formula

One quantity that should, according to the discussions above, vary only smoothly with mass number is the binding energy per particle. In fact, the total binding energy $B(Z, N)$, where N and Z are the number of neutrons and protons, respectively, grows with the number of nucleons, A, in such a way that the binding energy per particle $B(N, Z)/A$ stays fairly constant for nuclei with more than twelve nucleons:

$$\left.\frac{B(N,Z)}{A}\right|_{A>12} \simeq -8.5\ [\text{MeV/nucleon}]. \tag{1.1}$$

The explanation for the binding energy per nucleon being approximately constant can be given in the following qualitative way: If the binding of a system of A particles comes from the interaction energy of all possible two-particle pair combinations, then the total binding energy should grow roughly like the number of all possible pairs, namely as $\frac{1}{2} \cdot A \cdot (A-1)$. The binding energy per particle should then be proportional to A. This is the case, for example, for the binding energies of the electrons in an atom.

The completely different behavior of the nuclear binding energy can be attributed to the saturation property of the nuclear forces: one nucleon in the nucleus interacts with only a limited number of nucleons. This has its origin in the short-range nuclear force and the combined effect of the Pauli and uncertainty principles: The total binding energy is a subtle difference effect between the total kinetic energy and the total potential energy; whereas due to the Pauli and uncertainty principles and the hard core [Br 65a], the kinetic energy rises steeply to positive values at decreasing nucleon distances [BM 69, p. 252] (see Fig. 1.1), and the potential energy is lowered to more and more negative energies. For small enough ($d \lesssim 2$ fm) internucleon distances the kinetic energy takes over so that the total

binding energy becomes positive, whereas for large distances ($d \lesssim 3$ fm), there is hardly any potential energy because of the short-range nucleon–nucleon force. Finally, a shallow minimum develops at around 2.4 fm, which is, compared to the strength of the nuclear forces, quite small. From these considerations it becomes clear why there can be only a limited number of nucleons within the interaction range of one nucleon. In this argument we have not taken into account surface effects and the Coulomb force. They have to be treated separately, as we shall see. Neglecting these effects, we expect that the total binding energy rises linearly with A, as in Eq. (1.1).

The saturation property also explains qualitatively the features found experimentally (by, for instance, electron scattering, μ-mesonic x-rays, etc. [FN 76]), that is, the roughly constant density of nucleons inside the nucleus and the nucleus's relatively sharp surface. Describing the nucleus as a sphere with a constant density and a *sharp* surface, we get for its radius

$$R = r_0 A^{1/3}, \tag{1.2}$$

where the parameter r_0 has empirically the value

$$r_0 = 1.2\ [\mathrm{fm}]. \tag{1.3}$$

In Fig. 1.2, we see the dependence of the binding energy per nucleon on the mass number A in more detail.

Up to about $A = 12$, we get a steep rise until $B/A \simeq 8$ MeV; at $A \simeq 60$, we obtain the maximum of a little less than 9 MeV. The binding energy per

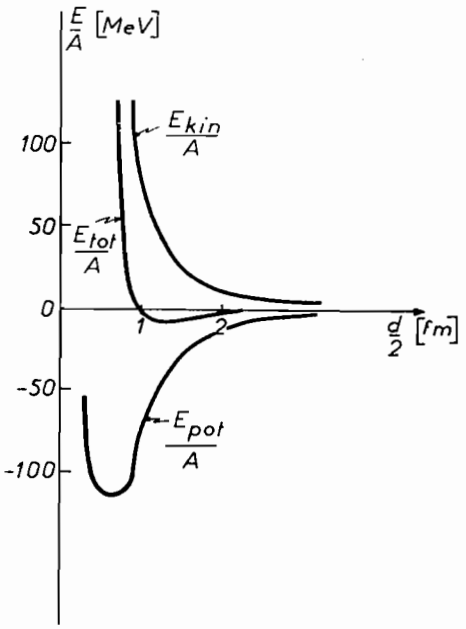


Figure 1.1. Qualitative explanation of nuclear saturation as a subtle difference effect of E_{kin} and E_{pot}.

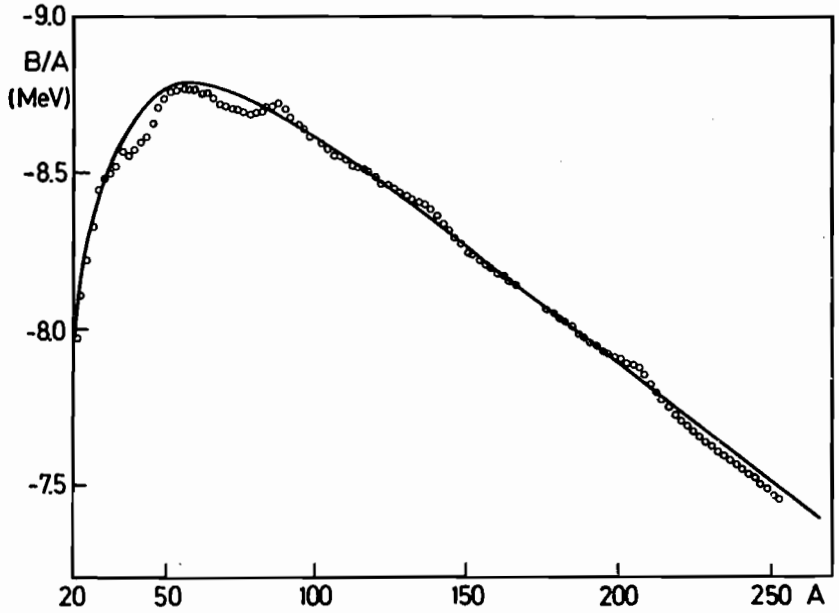


Figure 1.2. Experimental values of B/A for β-stable odd-A(o) nuclei and the calculated curve using a mass formula similar to Eq. (1.4). (From [Ho 75].)

nucleon then drops slightly until at $A=250$ it is about 7.5 [MeV]. This is due to the increasing influence of the Coulomb repulsion of the protons.

There have been many attempts to reproduce the behavior of B/A as a function of N and Z. The best known formula of this kind is the semi-empirical mass formula of Bethe and Weizsäcker [We 35, BB 36]

$$B(N,Z)=a_V A+a_S A^{2/3}+a_C\frac{Z^2}{A^{1/3}}+a_I\frac{(N-Z)^2}{A}-\delta(A), \qquad (1.4)$$

where one obtains by a fit (see also [MS 66, My 69, MS 69]):

$$a_V=-15.68; \quad a_S=18.56; \quad a_C=0.717; \quad a_I=28.1 \quad [\text{MeV}]$$

$$\delta(A)=\left\{\begin{array}{ll} 34\cdot A^{-3/4} & \text{for even–even} \\ 0 & \text{for even–odd} \\ -34\cdot A^{-3/4} & \text{for odd–odd} \end{array}\right\} \text{nuclei.} \qquad (1.5)$$

In Fig. 1.2 we see that one obtains quite good overall agreement with the experimental curve with these kinds of semi-empirical mass formulae.

The physical meaning of formula (1.4) is the following. The first term is usually called the *volume term*, because it is proportional to A [$\propto R^3$ with Eq. (1.2)]. The reason that the value of a_V is not -8.5 [MeV] as in Eq. (1.1) is the following. In order to get the value -8.5, one has to average over a wide range of A, and the other terms (a_S, a_C, and a_I) are positive and not negligible.

The second term is proportional to $A^{2/3}$ ($\propto R^2$), and is therefore called the *surface term*. It results from the fact that the nucleons close to the surface contribute less to the total binding energy. One can calculate from

the parameter a_S a surface tension coefficient σ defined as the surface energy per unit area, and get with Eq. (1.3):

$$\sigma = \frac{a_S}{4\pi r_0^2} = 1.03\ [\mathrm{MeV \cdot fm^{-2}}]. \tag{1.6}$$

The third term takes into account the Coulomb repulsion of the protons. It can be calculated approximately by assuming the charges to be uniformly distributed over a sphere. The *Coulomb energy* of such a system is proportional to the number of proton pairs ($\propto Z^2$) and inversely proportional to the radius.

Since the protons repel one another, it would be energetically more favorable for a nucleus to have only neutrons—if there were no Pauli principle. A proton decaying into a neutron must enter a state above the neutron Fermi level (see Chap. 2), which is energetically unfavored. The energy balance of the neutron excess $N-Z$ is taken care of in the fourth term of Eq. (1.4), the so-called *symmetry energy*. It cannot depend on the sign of $N-Z$. In the Fermi gas model [SF 74, p. 127 f], one can show that it is proportional to $(N-Z)^2/A$. The quadratic dependence of the binding energy on the proton–neutron mass difference is experimentally very well confirmed. Only the base of the experimental parabola is different according to whether we are considering an even–even, even–odd or odd–odd nucleus. This is due to the so-called *pairing effect*, as we shall see in Chapter 6, and is taken care of by the last term in Eq. (1.4).

Some aspects of the semi-empirical mass formula will be discussed again in Chapter 13 in the context of the Thomas–Fermi approach to nuclear physics.

It should be noted that Eq. (1.4) gives only an overall smooth fit to the binding energy as a function of A, and that locally there are strong deviations from it (see Fig. 2.2), mostly due to shell effects, which will be discussed in Section 2.9.

1.3 Deformation Parameters

Up to now, we have only studied static properties of the liquid drop model. In the following, we will assume that the nucleus has a sharp surface* which must not necessarily be spherical, and we imagine it to undergo dynamical shape or surface oscillations.

Before we can investigate these oscillations, we have to parametrize the surface in some way. One possibility is to describe it by the length of the radius vector pointing from the origin to the surface

$$R = R(\theta, \phi) = R_0\left(1 + \alpha_{00} + \sum_{\lambda=1}^{\infty} \sum_{\mu=-\lambda}^{\lambda} \alpha_{\lambda\mu}^{*} Y_{\lambda\mu}(\theta, \phi)\right) \tag{1.7}$$

where R_0 is the radius of the sphere with the same volume. Such a surface

*Myers and Swiatecki [MS 69, 73] have given up this assumption and introduced a refined liquid drop model with a diffuse surface, the so-called "droplet model" [Ni 72].

is certainly not the most general one* but it is widely used and extremely useful for problems of nuclear structure.

The constant α_{00} describes changes of the nuclear volume. Since we know that the incompressibility of the nuclear fluid is rather high, we require that the volume be kept fixed for all deformations as

$$V=\frac{4}{3}\pi R_0^3. \tag{1.8}$$

This defines the constant α_{00}. Up to second order, we get [EG 70]

$$\alpha_{00}=-\frac{1}{4\pi}\sum_{\lambda>1,\ \mu}|\alpha_{\lambda\mu}|^2. \tag{1.9}$$

The term $\lambda=1$ describes mainly (at least for small deformations) a translation of the whole system. The three parameters $\alpha_{1\mu}$ can be fixed by the condition that the origin coincides with the center of mass

$$\int_V \mathbf{r}\, d^3r=0. \tag{1.10}$$

If the expansion (1.7) contains only even values of λ, this is fulfilled automatically. Otherwise $\alpha_{1\mu}$ starts with second order in the $\alpha_{\lambda\mu}$ ($\lambda>2$). Therefore, in the following we omit both α_{00} and $\alpha_{1\mu}$, since we shall restrict our discussion to small deformations.

It is instructive to look at the shapes of lowest multipolarity in the expansion (1.7), as displayed in Fig. (1.3). The deformations corresponding

*For instance, shapes of two separated fragments in the fission process cannot be represented by (1.7), since R is then multivalued.

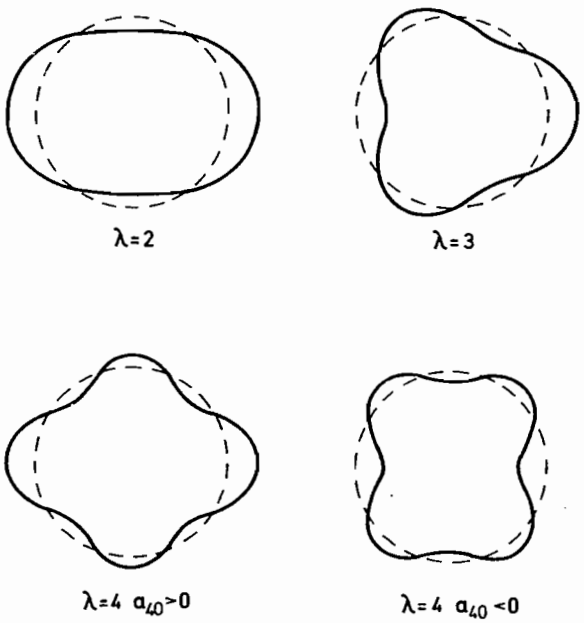


Figure 1.3. Nuclear shapes with quadrupole ($\lambda=2$), octupole ($\lambda=3$), and hexadecupole ($\lambda=4$) deformations.

to $\lambda=2$ look like ellipsoidal deformations. It should be noticed, however, that this is true only up to first order. A pure ellipsoid has non-vanishing $\alpha_{\lambda\mu}$ for all $\lambda \geqslant 2$.

Another condition on R, and therefore on the parameters $\alpha_{\lambda\mu}$, is the fact that R should be invariant under a reflection of the coordinate system and under a rotation of the coordinate system. In order for this to be the case, the $\alpha_{\lambda\mu}$ must be multiplied by a factor $(-)^{\lambda}$ under a parity transformation, and must behave like $Y_{\lambda\mu}(\theta, \phi)$ under a rotation of the coordinate system (characterized by the Euler angles $\Omega=(\alpha, \beta, \gamma)$ [Ed 57, Eq. (5.2.1.)]), i.e.,

$$(Y_{\lambda\mu})_{\text{new}} = \sum_{\mu'} D^{\lambda}_{\mu'\mu}(\Omega)(Y_{\lambda\mu'})_{\text{old}}, \tag{1.11}$$

$$a_{\lambda\mu} = \sum_{\mu'} D^{\lambda}_{\mu'\mu}(\Omega)\alpha_{\lambda\mu'},$$

where $D^{\lambda}_{\mu'\mu}(\Omega)$ are the Wigner functions of the rotation and $a_{\lambda\mu}$ are the deformation parameters in the new system.

To make sure that the radius R in Eq. (1.7) is real, we have to use the property $Y^{*}_{\lambda\mu}=(-)^{\mu}\cdot Y_{\lambda-\mu}$, and get

$$\alpha^{*}_{\lambda\mu}=(-)^{\mu}\alpha_{\lambda-\mu}. \tag{1.12}$$

This will turn out to be the time reversal behavior of the $\alpha_{\lambda\mu}$'s.

Before discussing the surface oscillations of general multipolarity, we mention two special cases:

(i) First are the axially symmetric deformations. Choosing the z-axis as symmetry axis, we find that $\alpha_{\lambda\mu}$ vanishes except when $\mu=0$. The deformation parameters $\alpha_{\lambda 0}$ are usually called β_{λ}.

(ii) In the case of quadrupole deformations ($\lambda=2$), we have five parameters $\alpha_{\lambda\mu}$. Not all of them describe the shape of the drop. Three determine only the orientation of the drop in space, and correspond to the three Euler angles. By a suitable rotation, we can transform to the body-fixed system characterized by three axes 1, 2, 3, which coincide with the principal axes of the mass distribution of the drop. The five coefficients $\alpha_{2\mu}$ reduce to two real independent variables a_{20} and $a_{22}=a_{2-2}$ ($a_{21}=a_{2-1}=0$), which, together with the three Euler angles, give a complete description of the system. It is convenient to introduce instead of a_{20} and a_{22} the so-called Hill–Wheeler [HW 53] coordinates β, γ ($\beta>0$) through the relation

$$a_{20}=\beta\cdot\cos\gamma,$$

$$a_{22}=\frac{1}{\sqrt{2}}\cdot\beta\cdot\sin\gamma, \tag{1.13}$$

from which we have

$$\sum_{\mu}|\alpha_{2\mu}|^2=a_{20}^2+2a_{22}^2=\beta^2 \tag{1.14}$$

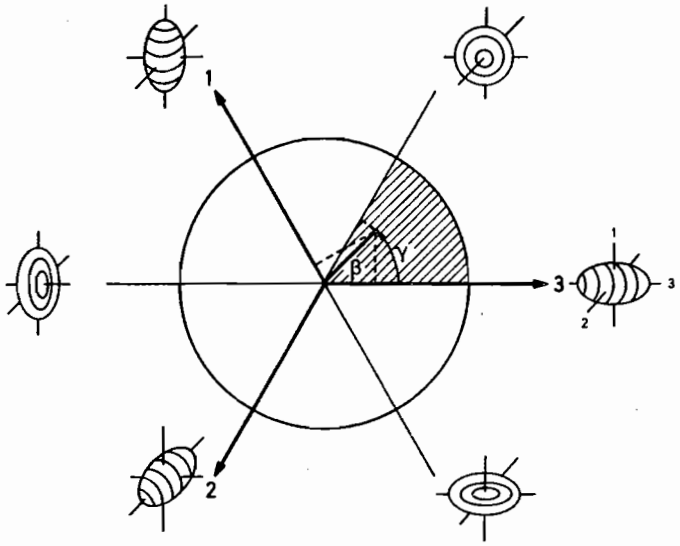


Figure 1.4. Nuclear shapes in the β, γ plane. The projections onto the three axes are proportional to the increments of δR_κ (1.16).

and

$$R(\theta,\phi)=R_0\Big\{1+\beta\sqrt{\frac{5}{16\pi}}\ \big(\cos\gamma(3\cos^2\theta-1) + \sqrt{3}\sin\gamma\sin^2\theta\cos 2\phi\big)\Big\}. \tag{1.15}$$

In Fig. (1.4) the $\lambda=2$ shapes are represented in the polar coordinates β, γ. We see that

(i) γ values of 0°, 120°, and 240° yield prolate spheroids with the 3, 1 and 2 axes as axes of symmetry;
(ii) $\gamma=180°$, 300°, and 60° lead to the corresponding oblate shapes
(iii) when γ is not a multiple of 60° it corresponds to a triaxial shape;
(iv) there are discrete symmetries, namely, one can interchange all three axes without changing the shape, which means an invariance under the point group D_2. The interval $0<\gamma<60°$ is sufficient to describe all the $\lambda=2$ shapes. All other points in Fig. 1.4 are obtained by suitable exchanges of the different axes.
(v) We can calculate the increments of the three semi-axes in the body-fixed frame as functions of β and γ:

$$\delta R_1=R\Big(\frac{\pi}{2},0\Big)-R_0=R_0\sqrt{\frac{5}{4\pi}}\ \beta\cos\Big(\gamma-\frac{2\pi}{3}\Big),$$

$$\delta R_2=R\Big(\frac{\pi}{2},\frac{\pi}{2}\Big)-R_0=R_0\sqrt{\frac{5}{4\pi}}\ \beta\cos\Big(\gamma+\frac{2\pi}{3}\Big),$$

$$\delta R_3=R(0,0)-R_0=R_0\sqrt{\frac{5}{4\pi}}\ \beta\cos\gamma,$$

or

$$\delta R_\kappa = R_0 \sqrt{\frac{5}{4\pi}}\ \beta \cos\left(\gamma - \frac{2\pi}{3}\kappa\right), \qquad \kappa = 1, 2, 3. \tag{1.16}$$

We have to remember, however, that the parameters β and γ (with (1.7)) only describe exactly ellipsoidal shapes in the limit of small β-values (see also Eq. 1.88).

1.4 Surface Oscillations About a Spherical Shape

The first kind of excitations are dynamical shape, or surface, oscillations. The dynamical variables are in this case the parameters which describe the surface, i.e., the surface coordinates $\alpha_{\lambda\mu}$ ($\lambda \geqslant 2$) of Eq. (1.7). They are considered to be functions of time: $\alpha_{\lambda\mu}(t)$. For the low-lying excitations one can expect that they produce small oscillations around the spherical equilibrium shape with $\alpha_{\lambda\mu} = 0$, and that the classical Hamilton function H_{coll} that describes this process is of a harmonic oscillator form [Bo 52]:

$$H_{\text{coll}} = T + V = \frac{1}{2} \sum_{\lambda,\ \mu} \left\{ B_\lambda |\dot{\alpha}_{\lambda\mu}|^2 + C_\lambda |\alpha_{\lambda\mu}|^2 \right\}, \tag{1.17}$$

Here the parameters of inertia B_λ and of stiffness C_λ are real constants. This is, in fact, the only quadratic form which is invariant under rotation and time reversal.*

Following the usual rules of canonical quantization (see for instance, [EG 70, p. 40]), we obtain the quantized form† (see also Appendix C)

$$\hat{H}_{\text{coll}} = \sum_{\lambda\mu} \hbar\Omega_\lambda \left(B^+_{\lambda\mu} B_{\lambda\mu} + \frac{1}{2} \right) \tag{1.18}$$

with the frequencies

$$\Omega_\lambda = \left(\frac{C_\lambda}{B_\lambda} \right)^{1/2}. \tag{1.19}$$

The operators $B_{\lambda\mu}$ obey Bose commutation rules

$$\left[B_{\lambda\mu}, B_{\lambda'\mu'} \right] = 0; \qquad \left[B_{\lambda\mu}, B^+_{\lambda'\mu'} \right] = \delta_{\lambda\lambda'} \delta_{\mu\mu'} \tag{1.20}$$

and have a corresponding Bose vacuum $|0)$ such that $B_{\lambda\mu}|0) = 0$. The Boson operators $B_{\lambda\mu}$ are related to the coordinates $\hat{\alpha}_{\lambda\mu}$ and corresponding momenta $\hat{\pi}_{\lambda\mu}$ by

$$\hat{\alpha}_{\lambda\mu} = \left(\frac{\hbar}{2 B_\lambda \Omega_\lambda} \right)^{1/2} \left(B^+_{\lambda\mu} + (-)^\mu B_{\lambda -\mu} \right), \tag{1.21}$$

$$\hat{\pi}_{\lambda\mu} = i \left(\frac{\hbar}{2} B_\lambda \Omega_\lambda \right)^{1/2} \left((-)^\mu B^+_{\lambda -\mu} - B_{\lambda\mu} \right).$$

*The time reversal operation is discussed in great detail by Messiah [Me 61] and it is shown that in a system without spin, as we have here, time reversal corresponds to complex conjugation.

†One should not mix up the boson operators $B_{\lambda\mu}$ with the inertia parameters B_λ.

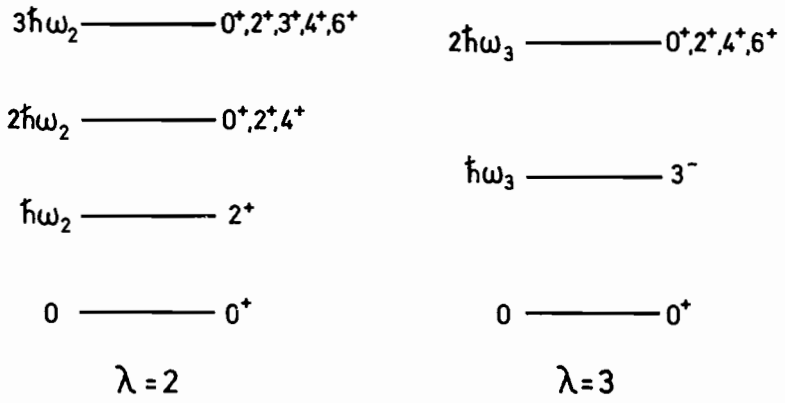


Figure 1.5. Harmonic energy spectra for the quadrupole ($\lambda=2$) and octupole ($\lambda=3$) surface oscillations.

From the above considerations, it follows that for each λ we have a harmonic spectrum of surface vibrations as illustrated in Fig. 1.5.

From the fact that $\alpha_{\lambda\mu}$ and $B^+_{\lambda\mu}$ behave like spherical tensors under rotations of the coordinate system (Eq. (1.11), [Ed 57, Eq. (5.2.1)]) we know the commutation relations of the angular momentum operators with $B^+_{\lambda\mu}$ and find that the one-boson states

$$|\lambda\mu)=B^+_{\lambda\mu}|0) \tag{1.22}$$

have angular momentum $I=\lambda$ and z-component $M=\mu$ with parity $(-)^\lambda$.

To construct multi-boson states we have to use the rules of angular momentum coupling [Ed 57] and also have to take into account that states with more than two bosons are symmetric under the exchange of any two of them. For instance, we get, for the superposition of two quadrupole bosons ($\lambda=2$), the three combinations $I^\pi=0^+,2^+,4^+$

$$|IM)=\frac{1}{\sqrt{2}}\sum_{\mu_1\mu_2} C^{2\,2\,I}_{\mu_1\mu_2 M} B^+_{2\mu_1} B^+_{2\mu_2}|0). \tag{1.23}$$

The states with $I=1,3$ vanish identically because of the behavior of the Clebsch–Gordan coefficients* [Ed 57, Eq. (3.5.14)]

$$C^{2\,2\,I}_{\mu_1\mu_2 M}=(-)^I C^{2\,2\,I}_{\mu_2\mu_1 M}.$$

under an exchange of μ_1 and μ_2.

Indeed, many spherical nuclei show in their spectrum a low-lying 2^+ state and, at roughly double the excitation energy, a so-called two-boson triplet $(0^+,2^+,4^+)$ which is, however, usually split up a little (see Fig. 9.4).

The constants B_λ and C_λ can be calculated within the fluid picture; they depend on the flow associated with the surface oscillations. Therefore, it is necessary to postulate the nature of the fluid motion within the drop. At this point we should again discuss what the concept of a nuclear fluid

* We use the symbol $C^{j_1 j_2 j_3}_{m_1 m_2 m_3}$ which is the same as $(j_1 m_1 j_2 m_2 | j_1 j_2 j_3 m_3)$.

implies. In Chapter 13 we will investigate this point in detail and show that the fact that the nucleus is a Fermi liquid and not an ordinary liquid plays an important role. The simplest assumption about the flow pattern of the fluid we can make is that it is irrotational, i.e., $\mathbf{rot}\,\mathbf{v}(\mathbf{r})=0$, where $\mathbf{v}$ is the velocity field. We shall also study the justification of this point in Chapter 13, but for the moment let us take it for granted and thus have:

$$\mathbf{v}(\mathbf{t})=-\nabla\Phi(\mathbf{r}). \tag{1.24}$$

The next assumption is that of *incompressibility*, which is quite well justified for nuclei. It means that the density inside the nucleus is constant ($\dot{\rho}=0$), and we get from the equation of continuity

$$\nabla\mathbf{v}=0 \tag{1.25}$$

and, from (1.24),

$$\Delta\Phi=0. \tag{1.26}$$

The most general solution of Eq. (1.26) regular at the origin can be written in the form

$$\Phi(\mathbf{r})=\sum_{\lambda\mu} d^{*}_{\lambda\mu} r^{\lambda} Y_{\lambda\mu}(\theta,\phi). \tag{1.27}$$

For small deformations we have the boundary condition that the radial component of the velocity is, in lowest order, given by:

$$v_r=-\frac{\partial}{\partial r}\Phi=\dot{R} \qquad \text{at} \quad r=R_0,$$

which, with Eq. (1.7), yields the following relation between the coefficients $d_{\lambda\mu}$ and $\alpha_{\lambda\mu}$.

$$d_{\lambda\mu}=-\frac{1}{\lambda}R_0^{2-\lambda}\dot{\alpha}_{\lambda\mu}. \tag{1.28}$$

The kinetic energy of the surface vibrations is given by

$$T=\frac{m}{2}\rho\int_V \mathbf{v}^2(\mathbf{r})\,d^3r=\frac{m}{2}\rho\int_V|\nabla\Phi|^2\,d^3r=\frac{m}{2}\rho\oint_S\Phi^{*}\nabla\Phi\,ds,$$

where ρ is the constant density of nucleons with the mass m. Using the gradient formula for spherical harmonics [Ed 57, Eq. (5.9.17)] and [Ed 57, Eqs. (5.9.13) and (5.9.16)] we arrive in the approximation of small deformations (integrating over a sphere), in the following expression for the kinetic energy.

$$T=\frac{R_0^5 m\rho}{2}\sum_{\lambda\mu}\frac{|\dot{\alpha}_{\lambda\mu}|^2}{\lambda}. \tag{1.29}$$

Comparison with (1.17) yields the mass parameter

$$B_\lambda=\frac{\rho m R_0^5}{\lambda}=\frac{3}{4\pi\lambda}A\cdot mR_0^2. \tag{1.30}$$

For oscillations about a spherical equilibrium shape the mass parameter is not a function of μ; this would not be so for a deformed nuclear drop.

The potential energy of a liquid drop with a surface deformation characterized by the parameters $\alpha_{\lambda\mu}$ can be obtained from the coefficients of the Bethe-Weizsäcker formula (1.4) if we neglect changes of the symmetry and pairing energy with deformation.

Because of the assumption of incompressibility, it is tempting to say that the volume term does not depend on the deformation. This is, however, only true for ordinary fluids, and we will see in Chapter 13 how in quantum fluids the volume term can depend on α in a quite subtle fashion. In the usual treatment of the liquid drop model [BM 53, EG 70], however, the volume term is not taken into account, and therefore the deformation energy has only two parts, resulting from the surface and Coulomb terms in Eq. (1.4). As we will discuss in more detail in Chapter 13 this will be sufficient for the monopole and the dipole resonance but not for resonances of other multipolarities.

The deformation energy is defined as the difference between the energy of the deformed and spherical drop:

$$V(\alpha)=E_S(\alpha)-E_S(0)+E_C(\alpha)-E_C(0). \tag{1.31}$$

The surface energy is given by the product of the surface with the surface tension σ [Eq. (1.6)]. With techniques similar to those used in the derivation of Eq. (1.29), we find up to second order in $\alpha_{\lambda\mu}$ [Wi 64, Chap. 2]:

$$E_S(\alpha)=\sigma\oint_S ds=E_S(0)+\frac{1}{2}\sum_{\lambda\mu}(\lambda-1)(\lambda+2)R_0^2\sigma|\alpha_{\lambda\mu}|^2. \tag{1.32}$$

The Coulomb energy E_C is the sum of interactions between pairs of volume elements d^3r_1 and d^3r_2 [Wi 64]

$$E_C(\alpha)=(Ze)^2\int_V\int\frac{d^3r_1d^3r_2}{|\mathbf{r}_1-\mathbf{r}_2|}=E_C(0)-\frac{1}{2}\sum_{\lambda\mu}\frac{3(\lambda-1)(Ze)^2}{2\pi(2\lambda+1)R_0}|\alpha_{\lambda\mu}|^2. \tag{1.33}$$

From (1.32), (1.33), and (1.17), we thus get the stiffness coefficients ($\lambda\geqslant 2$):

$$C_\lambda=(\lambda-1)(\lambda+2)R_0^2\sigma-\frac{3(\lambda-1)}{2\pi(2\lambda+1)}\frac{(Ze)^2}{R_0}. \tag{1.34}$$

In principle, we are now able to calculate nuclear spectra from Eq. (1.19) and the coefficients B_λ and C_λ. It turns out, however, that the reproduction of spectra is not the most sensitive test for a nuclear model. A quantum mechanical state is represented by a wave function. Electromagnetic moments and transition probabilities depend strongly on the wave functions and provide a much better test. We therefore first discuss such quantities before comparing the theory with experimental data.

In Appendix B the calculation of the electromagnetic properties of a nucleus is shown. The essential quantities are the electric and magnetic multipole operators.

The electric multipole operators are in the limit of long wavelengths (low transition energies) given by [Eq. (B.18)]

$$\hat{Q}_{\lambda\mu}=e\int_V\rho_p(\mathbf{r})r^\lambda Y_{\lambda\mu}(\theta,\phi)\,d^3r.$$

We can express them by the cordinates $\alpha_{\lambda\mu}$, taking in the integral a constant proton density ρ_p for a shape defined by Eq (1.7) and get

$$\hat{Q}_{\lambda\mu}=\rho_p\cdot e\cdot\int_{4\pi} d\cos\theta\, d\phi\, Y_{\lambda\mu}(\theta,\phi)\frac{1}{\lambda+3}R^{\lambda+3}(\theta,\phi),$$

which is up to second order in $\alpha_{\lambda\mu}$ (with [Ed 57, Eq. (4.6.3)])

$$\hat{Q}_{\lambda\mu}=\frac{3e}{4\pi}ZR_0^{\lambda}\Bigg\{\hat{\alpha}_{\lambda\mu}+\frac{1}{2}(\lambda+2)\sum_{\lambda_1\mu_1}\sum_{\lambda_2\mu_2}\hat{\alpha}_{\lambda_1\mu_1}\hat{\alpha}_{\lambda_2\mu_2}(-)^{\mu}$$

$$\cdot\sqrt{\frac{(2\lambda_1+1)(2\lambda_2+1)(2\lambda+1)}{4\pi}}\begin{pmatrix}\lambda_1 & \lambda_2 & \lambda\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}\lambda_1 & \lambda_2 & \lambda\\ -\mu_1 & -\mu_2 & \mu\end{pmatrix}\Bigg\}. \quad (1.35)$$

The magnetic multipole operators are given by [Eq. (B. 22)]

$$\hat{M}_{\lambda\mu}=\frac{1}{c(\lambda+1)}\int_V(\mathbf{r}\times\mathbf{j}(\mathbf{r}))\left(\nabla r^{\lambda}Y_{\lambda\mu}\right)d^3r.$$

As in the case of the mass parameters, we could again use the assumption of irrotational flow to define the current density $\mathbf{j}$ and express $\hat{M}_{\lambda\mu}$ by $\alpha_{\lambda\mu}$ and $\dot{\alpha}_{\lambda\mu}$ (see [Da 68, Chap. 6]). However, we will restrict ourselves to the case of $M1$ operators, which form a vector

$$\hat{\mathbf{M}}1=\sqrt{\frac{3}{4\pi}}\;\hat{\boldsymbol{\mu}}.$$

Since we have no spins in the system, we get for the magnetic dipole moment

$$\hat{\boldsymbol{\mu}}=\frac{1}{2c}\int_V(\mathbf{r}\times\mathbf{j})\,d^3r=\frac{Z}{A}\frac{e}{2mc}\int_V(\mathbf{r}\times m\mathbf{v})\,d^3r=\frac{Z}{A}\frac{e\hbar}{2mc}\hat{\mathbf{I}}=g_R\hat{\mathbf{I}}\mu_N \quad (1.36)$$

with the gyromagnetic ratio of the rotor

$$g_R=\frac{Z}{A}. \quad (1.37)$$

The calculation of lifetimes and transition probabilities requires the knowledge of $BE\lambda$- and $BM\lambda$-values [see Eq. (B. 73)] defined by

$$B\left(\left\{\begin{matrix}E\\M\end{matrix}\right\}\lambda, I_i\to I_f\right)=\frac{1}{2I_i+1}\left|<I_f\|\left\{\begin{matrix}\hat{Q}_{\lambda}\\ \hat{M}_{\lambda}\end{matrix}\right\}\|I_i>\right|^2. \quad (1.38)$$

Since the $M1$-operator (1.36) conserves angular momentum, $M1$-transitions are forbidden in this model. The most important transitions are $E2$-transitions.

As an example, we calculate the $BE2$-value for the transition from the one-boson state $B^+_{2\mu}|0)$ to the ground state $|0)$. Expressing $\hat{\alpha}_{2\mu}$ by the operators $B_{2\mu}$ and $B^+_{2\mu}$ (1.21), and using the wave function (1.22), we get to first order in the $\alpha_{2\mu}$'s

$$B(E2, 2_1^+\to 0^+)=\left(\frac{3}{4\pi}ZeR_0^2\right)^2\frac{\hbar}{2B_2\Omega_2}. \quad (1.39)$$

Before we compare these theoretical results of our model with experimental data, we must discuss which levels in the excitation spectra of nuclei would be appropriate candidates for such surface vibrations. We restrict the following discussion to 2^+ states. For other angular momenta similar considerations apply.

Figure 1.6 shows schematically the structure of the 2^+ spectra. They have a discrete part and a continuum with resonances. Among the discrete lines one 2^+ level is usually very low in energy. With a few exceptions it is the lowest excited state in each nucleus and, as shown in Fig. 1.6, it carries a large $BE2$ value, i.e., it has a high transition probability to the ground state (see Appendix B). The measured $BE2$ values are for spherical nuclei roughly ten to twenty times larger than one would expect from a pure single-particle transition [Weisskopf unit, see Eq. (B. 85)].

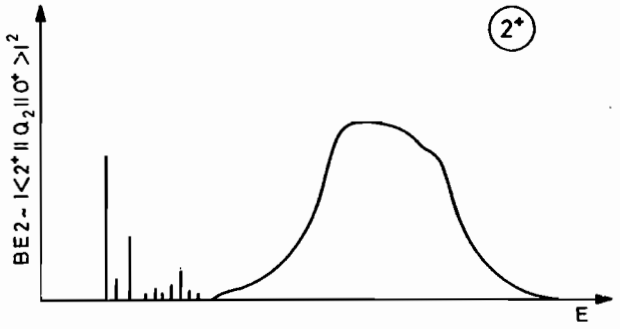


Figure 1.6. Schematic representation of the 2^+ spectra in nuclei. The ordinate gives the $BE2$ values for the discrete levels and the density of the $BE2$ strength in the resonance region. These quantities measure the transition probability to the ground state. The units are arbitrary.

The low-lying 2^+ states therefore have collective character, i.e., many particles contribute and they have very often been interpreted as surface quadrupole vibrations. Figure (1.7) shows the energy E_{2^+} for the lowest 2^+ state in even–even nuclei. One observes large shell effects (see Chap. 2). Only the average trend is given by the liquid drop model with irrotational flow [Eqs. (1.19) (1.30) and (1.34)]. The absolute value is off by a factor of five. The reason for this failure will become clear below.

Experimentally, it has been found that there is a strong correlation between the $BE2$ value of the first 2^+ state and its energy $E_{2_1^+} = \Omega_2$ [Gr 62]:

$$E_{2_1^+} B(E2, 2^+ \to 0^+) \simeq (25 \pm 8) \frac{Z^2}{A} \left[\text{MeV } e^2 \text{ fm}^4\right]. \tag{1.40}$$

This empirical relation holds for all the nuclei throughout the nuclear table. Equation (1.39) shows the same energy dependence, but the A dependence is different.The strong shell effects in the low-lying 2^+ states indicate that they cannot be pure quadrupole surface oscillations and that there are other states which also have the character of such collective vibrations. From Eqs. (1.22), (1.21), and (1.35) we see that the quadrupole

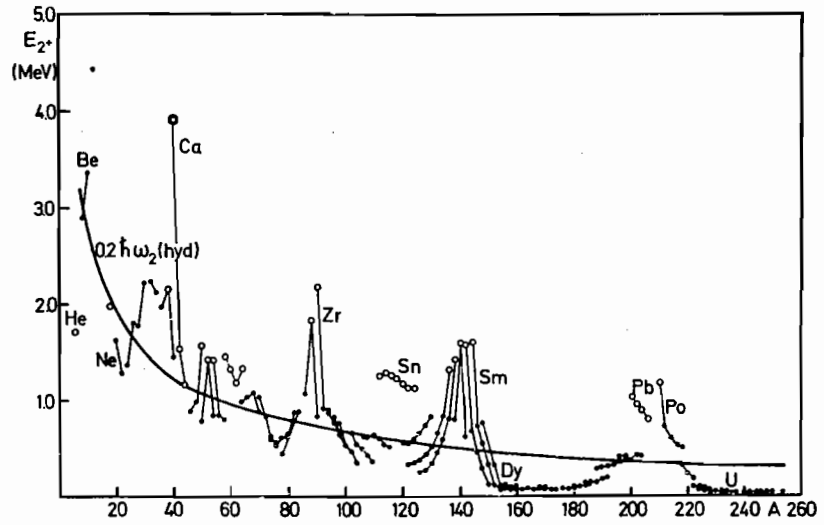


Figure 1.7. The energy of the first 2^+ state in even–even nuclei. The nuclei with closed neutron or proton shells are marked by open circles. (From [NN 65].)

surface vibration can be represented in linear order in α as:

$$|2\mu) = B^+_{2\mu}|0) \propto \hat{\alpha}_{2\mu}|0) \propto \hat{Q}_{2\mu}|0). \tag{1.41}$$

The overlap of an arbitrary state $|\nu\rangle$ with the quadrupole surface vibration is therefore proportional to its $BE2$ value:

$$|\langle\nu|2\mu\rangle|^2 \propto |\langle\nu|\hat{Q}_{2\mu}|0\rangle|^2 \propto B(E2, \nu \to 0), \tag{1.42}$$

and the probability that it can be interpreted as such a vibration is given by the percentage to which it exhausts the sum rule

$$S^0_{2\mu} = \sum_{\nu \neq 0} |\langle\nu|\hat{Q}_{2\mu}|0\rangle|^2. \tag{1.43}$$

Only if one state exhausts this sum rule to a large extent is it meaningful to call it a quadrupole surface vibration.

In Section 8.7 we will discuss in great detail the sum rules and how they can be evaluated. It is evident that in a model where the state $\hat{Q}_{\lambda\mu}|0\rangle$ is an eigenstate of the system, like the model we are now investigating, this state exhausts the sum rules completely, because all the other states are orthogonal to it.

Experimentally, it has been found that the low-lying 2^+ state usually exhausts about 10–20% of the sum rule. The major part is exhausted by the resonances in the continuum [see Fig. (1.6)], the so-called giant resonances.

Such giant resonances for different I values have been observed. The most famous is the giant dipole resonance (1^-) which has been well known since more than 30 years and lies at an energy (see Fig. 1.8) (for more details see Chap. 13):

$$\Omega^{GR}_{1^-} \simeq 77 \cdot A^{-1/3+-1/6}\,[\mathrm{MeV}]. \tag{1.44}$$

As we shall see in Chapters 8 and 13, it corresponds to a vibration of the neutron and proton sphere against one another and cannot be described in the present simple model.

In the last ten years, further giant resonances have been observed. The most important example is the isoscalar giant quadrupole resonance. It lies at (see Fig. 1.8)

$$\Omega_{2^+}^{GR} \simeq 62 \cdot A^{-1/3} \,[\mathrm{MeV}] \tag{1.45}$$

and exhausts in most cases a major part of the S_{2^+} sum rule.

It seems, therefore, to be more reasonable to interpret this resonance as the quadrupole surface vibration mode discussed so far. The liquid drop model in this form is, however, not able to give the proper A-dependence. From Eqs. (1.19), (1.30), and (1.34) we get

$$\Omega_2 \propto A^{-1/2},$$

which does not agree with the experimental value (1.45). In Chapter 13 we will see that the reason for this deviation comes from the fact that the potential energy coefficients C_λ correspond to the total binding energy of the liquid drop [(1.31), (1.34)]. This total binding energy is, however, a sum of intrinsic kinetic and potential energy. The fact that (at least, for small deformations) the intrinsic kinetic energy depends on the deformation has been neglected in (1.34). This can be understood simply as an effect of the long mean free path of the nucleons and the uncertainty principle, which states that in an ellipsoidal shape the nucleons along the short axis have higher momenta than those along the long axis. An ellipsoidal momentum distribution, however, yields a larger kinetic energy than a spherical one. Since all particles are affected, it is a volume effect which dominates in general over the surface dependence given in (1.34). In Chapter 13 we will give a detailed discussion of this point.

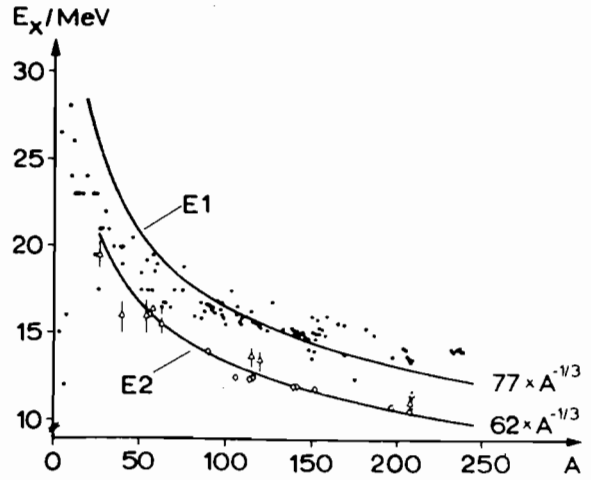


Figure 1.8. The energy of the giant dipole and the isoscalar giant quadrupole resonance as a function of mass number [Wa 73].

In spite of these deficiencies, the boson model presented so far has been extensively used over the years and has proved quite successful in explaining many features of the nuclear spectra [EG 70]. The reason for this is that many low-lying nuclear excitations have a rather collective character and can be represented to a good approximation as bosons. The parameters B_λ and C_λ may not be given very well in the liquid drop model, but, as we will shown in Chapters 8 and 9, there are microscopic theories for fermions which allow more reliable calculations of B_λ and C_λ; therefore, we should consider B_λ and C_λ more as open adjustable parameters than as determined by the LDM.

The microscopic theories also will show that the harmonic approximations (1.17) have only a very rough validity in the limit of very small vibrations. For collective motions with larger amplitudes, one has to take into account anharmonic terms such as

$$\alpha^4, \alpha^2 \cdot \dot{\alpha}^2, \dot{\alpha}^4, \ldots .$$

Again the corresponding parameters can be adjusted to experiment or calculated from a microscopic many-body theory (see Chap. 9).

1.5 Rotations and Vibrations for Deformed Shapes

1.5.1 The Bohr Hamiltonian

The pure liquid drop model has a stable equilibrium only for spherical surfaces. As we shall see later (Sec. 2.8) it can happen as a consequence of quantum mechanics—i.e., shell effects—that the potential $V(\alpha)$ in the collective Hamiltonian has minima at finite non-vanishing values of $\alpha = \alpha_0$. In such cases the nucleus can have a stable ground state deformation.

In this case, the nucleus can exhibit *rotations* which can be described by time-dependent surface parameters $\alpha_{\lambda\mu}$ in the laboratory frame. We shall call these rotations collective ones: This kind of rotation will not be possible around an axis of symmetry, because we cannot distinguish the rotated system from the original one in our variables $\alpha_{\lambda\mu}$.

In a quantum mechanical description, a system with an axis of symmetry (for example, the z-axis) is given by a wave function which is an eigenfunction of the angular momentum operator J_z, and any rotation about this axis produces only a phase. The rotating system has, therefore, the same wave function as the ground state, and the same energy.

This does not mean that there are no other degrees of freedom in the system that can be excited (for instance, single-particle degrees of freedom) and carry angular momentum parallel to the symmetry axis. Such a "rotation," however, we do not call collective rotation.

Since in almost all nuclei the quadrupole degree of freedom plays a fundamental role, we will restrict the following considerations to the case $\lambda = 2$.

Assuming that the nucleus has a stable ground state deformation, it is preferable to transform to the body-fixed system, defined by the principal axes of the mass distribution (as discussed in Sec. 1.3).

After this transformation, we have five dynamical variables Ω, β, γ* instead of the variables $\alpha_{2\mu}$ ($\mu=-2,-1,0,+1,+2$). We start again from the Hamiltonian (1.17). Only the potential in (1.17) is changed. It now has the form:

$$V(\beta,\gamma)=\tfrac{1}{2}C_{20}\left(a_{20}(\beta,\gamma)-a_{20}^0\right)^2+C_{22}\left(a_{22}(\beta,\gamma)-a_{22}^0\right)^2. \qquad (1.46)$$

This corresponds to a quadratic approximation in the vicinity of a deformed minimum β_0, γ_0. The idea is that the nucleus has this deformation in its ground state, and the excitations are rotations and small oscillations around this equilibrium deformation.

Microscopic calculations of potential landscapes show, for certain nuclei, well pronounced minima of V for finite values $\beta_0 \simeq 0.2-0.3$ and $\gamma_0=0$. These axially symmetric shapes are, therefore, the most important ones, and we will restrict a large part of our discussion to them.

The next step is the transformation of the kinetic energy in Eq. (1.17) to the body-fixed system. Applying Eq. (1.11), we have to differentiate the variables a_{20}, a_{22}, and Ω with respect to the time. Since the derivation is quite lengthy, and the intention in this chapter is not to be complete, we simply give the result (for the derivation, see, for example, [EG 70 Vol. I, Chap. 5]) for the so-called Bohr Hamiltonian [Bo 52]:

$$T=T_{\mathrm{rot}}+\tfrac{1}{2}B_2(\dot\beta^2+\beta^2\dot\gamma^2) \qquad (1.47)$$

with

$$T_{\mathrm{rot}}=\tfrac{1}{2}\sum_{\kappa=1}^{3}\vartheta_\kappa\omega_\kappa^2,$$

where ω_κ is the angular velocity around the body-fixed κ-axis and ϑ_κ are functions of β and γ given by

$$\vartheta_\kappa=4B_2\beta^2\sin^2\left(\gamma-\frac{2\pi}{3}\kappa\right), \qquad \kappa=1,2,3. \qquad (1.48)$$

In case we have fixed deformations β,γ, T_{rot} is the kinetic energy of a rotor with the moment of inertia ϑ_κ. As soon as we allow for changes of β and γ, the rotational and vibrational degrees of freedom are coupled by the deformation dependence of the moments of inertia. In this case, we no longer have a pure rotor. In fact, we see that in the case of $\beta_0=0$ the system can be transformed back to a harmonic vibrator (1.17) which has a harmonic spectrum.

However, even in the well deformed case with large stiffness parameters C and finite β_0 (where a constant $\beta=\beta_0$ is a rather good approximation),

*Unfortunately, the set of Euler angles $\Omega=(\alpha,\beta,\gamma)$ also contains the letters β and γ. However, we do not want to change this nomenclature [Ed 57] and we, as far as possible, use the abbreviation Ω.

$\mathcal{I}_\kappa$ are not the moments of inertia of a rigid rotor. Using (1.30), we get from Eq. (1.48) the so-called irrotational moment of inertia:

$$\mathcal{I}_\kappa^{\text{irr}} = \frac{3}{2\pi} mAR_0^2\beta^2 \sin^2\left(\gamma - \frac{2\pi}{3}\kappa\right), \qquad \kappa = 1, 2, 3.$$

It differs from the moment of inertia of a rigid body with the same deformation

$$\mathcal{I}_\kappa^{\text{rig}} = \frac{2}{5} mAR_0^2\left(1 - \sqrt{\frac{5}{4\pi}}\ \beta\cos\left(\gamma - \frac{2\pi}{3}\kappa\right)\right) \tag{1.49}$$

in the following ways.

(i) In the γ-dependence (Fig. 1.9), $\mathcal{I}^{\text{irr}}$ vanishes about the symmetry axes.

(ii) $\mathcal{I}^{\text{irr}}$ shows a strong dependence on the deformation ($\sim\beta^2$), whereas $\mathcal{I}^{\text{rig}}$ changes much less with β (its main part is the moment of inertia of a rigid sphere).

(iii) The experimental moment of inertia $\mathcal{I}^{\text{exp}}$ can be found from the energy of the first 2^+ state of a rotational band [see Eq. (1.64)] $\mathcal{I}^{\text{exp}} = 3/E_{2^+}$ [MeV^{-1}]. Applying the empirical rule (1.40) and the formula (1.74) for the $BE2$ value, we get a connection between the deformation parameter β and the moment of inertia

$$\mathcal{I}^{\text{exp}} = \frac{27}{80\pi^2} r_0^4 \frac{A^{4/3}\cdot\beta^2\cdot Z^2e^2}{E_{2^+}\cdot BE2} \approx \frac{27(1.2)^4\cdot\beta^2 A^{7/3}}{80\pi^2 25}$$

$$\approx \frac{\beta^2 A^{7/3}}{400}\ [\text{MeV}^{-1}]. \tag{1.50}$$

In the case of well deformed nuclei ($\beta\sim 0.2-0.4$), $\mathcal{I}^{\text{irr}}$ is usually smaller by a factor of 2–3 than the experimental values. On the other side, the values of $\mathcal{I}^{\text{rig}}$ are a factor of 2 too large:

$$\mathcal{I}^{\text{irr}} < \mathcal{I}^{\text{exp}} < \mathcal{I}^{\text{rig}}. \tag{1.51}$$

This shows that the flow structure within the nuclei is certainly not irrotational. On the other hand, it is not a rigid rotor, either.

The next step is again a quantization of the classical Hamiltonian (1.47). It is well known that there is no unique prescription for the quantization of a classical Hamiltonian in the general case [Me 61]. The ambiguity comes from the freedom in ordering noncommutable operators.* Commonly one adopts the Pauli prescription [Pa 33], which calculates the Laplace operator in curvilinear coordinates. If the classical kinetic energy has the form

$$T = \tfrac{1}{2}\sum_{ij}^{f} g_{ij}(\xi)\dot{\xi}_i\dot{\xi}_j, \tag{1.52}$$

* A discussion with a list of references on the quantization problem is found in [MD 73].

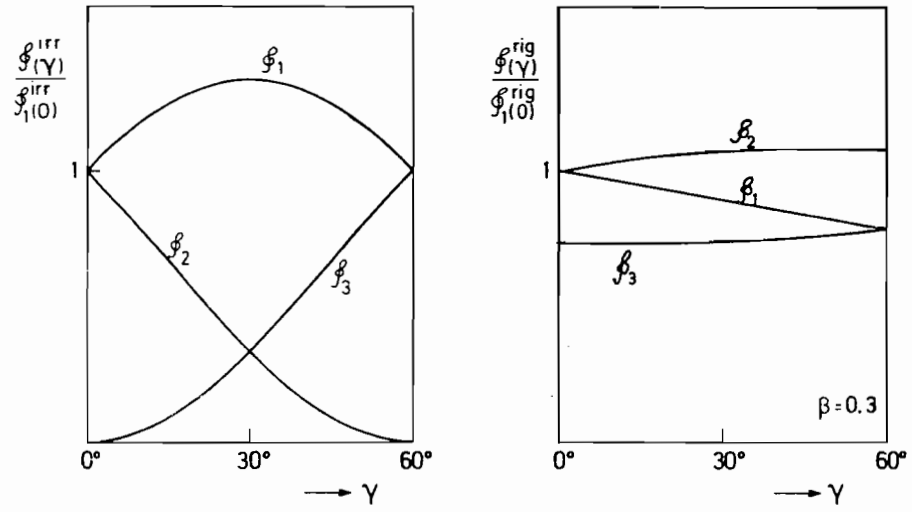


Figure 1.9. The γ-dependence of the irrotational ($\mathcal{J}^{irr}_{\kappa}$) and the rigid ($\mathcal{J}^{rig}_{\kappa}$) moments of inertia for fixed values of β.

then the corresponding quantized form is*

$$\hat{H}_{kin} = -\frac{\hbar^2}{2}\sum_{ij} g^{-1/2}\frac{\partial}{\partial\xi_i} g^{1/2}(g^{-1})_{ij}\frac{\partial}{\partial\xi_j}, \tag{1.53}$$

where g is the determinant and g^{-1} is the inverse of the matrix g_{ij}.

Applying this prescription to the Bohr Hamiltonians (1.46) and (1.47), we obtain:

$$\hat{H}_{coll} = \frac{-\hbar^2}{2B_2}\left[\beta^{-4}\frac{\partial}{\partial\beta}\left(\beta^4\frac{\partial}{\partial\beta}\right) + \frac{1}{\beta^2\sin 3\gamma}\frac{\partial}{\partial\gamma}\left(\sin 3\gamma\frac{\partial}{\partial\gamma}\right)\right] + \hat{T}_{rot} + V(\beta,\gamma), \tag{1.54}$$

where the rotational energy is found to be

$$\hat{T}_{rot} = \frac{\hat{I}_1^2}{2\mathcal{J}_1} + \frac{\hat{I}_2^2}{2\mathcal{J}_2} + \frac{\hat{I}_3^2}{2\mathcal{J}_3}. \tag{1.55}$$

The operators $\hat{I}_\kappa$ are the projections of the total angular momentum $\hat{\mathbf{I}}$ represented in the Euler angles onto the body-fixed axes (for details, see Appendix A). Figure (1.10) shows the total angular momentum and its components $\hat{I}_z = M$ and $\hat{I}_3 = K$. The eigenfunctions of $\hat{\mathbf{I}}^2, \hat{I}_z, \hat{I}_3$ are given by

$$|IMK\rangle = \sqrt{\frac{2I+1}{8\pi^2}}\; D^{I*}_{MK}(\Omega). \tag{1.56}$$

Since H_{coll}, $\hat{\mathbf{I}}^2$ and $\hat{I}_z$ commute, the eigenfunctions of the collective Hamil-

* To see that the Hamiltonian (1.53) is Hermitian, one has to take into account the volume element $\sqrt{g}\; d\xi_1 \ldots d\xi_f$.

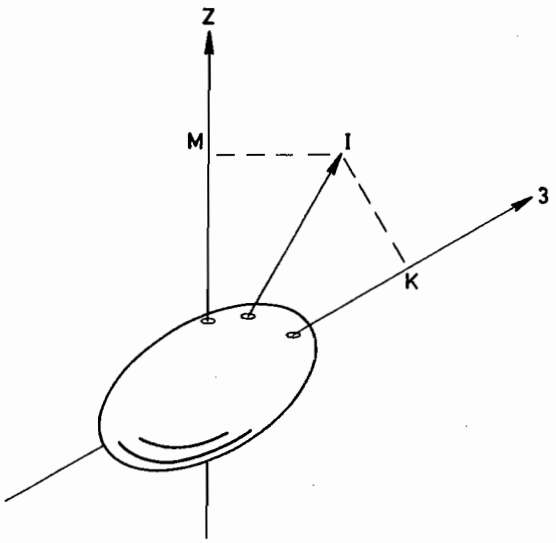


Figure 1.10. The relation between the total angular momentum **I** and its projection M onto the laboratory z axis and its projection K onto the body-fixed 3-axis.

tonian (1.54) have the general form

$$|\Psi^I_M\rangle = \sum_K g_K(\beta,\gamma)|IMK\rangle.$$

The triaxial rotor has certain discrete symmetries. $\hat{H}_{\text{coll}}$ is invariant under the point group D_2 [Bo 52]. Therefore, one classifies the eigenstates according to the irreducible representations of this group, and one can derive from this some properties of the spectra [Da 68].

One example is the rotation of 180° around the 1-axis,

$$\mathcal{R}_1 = e^{i\pi\hat{I}_1}, \tag{1.57}$$

which is equivalent to a reflection with respect to the 2, 3-plane together with a parity transformation. It commutes with $\hat{H}_{\text{coll}}$ and we require our eigenfunctions to have eigenvalue $+1$*:

$$\mathcal{R}_1|\Psi^I_M\rangle = |\Psi^I_M\rangle. \tag{1.58}$$

Using relation (A.24) of Appendix A, we find that this is only possible with

$$g_K(\beta,\gamma) = (-)^I g_{-K}(\beta,\gamma). \tag{1.59}$$

If we require, in the same way, symmetry with respect to

$$\mathcal{R}_2 = e^{i\pi\hat{I}_2}, \tag{1.60}$$

we get

$$g_K(\beta,\gamma) = (-)^{I+K} g_{-K}(\beta,\gamma). \tag{1.61}$$

* As we shall see in Eq. (11.123), the microscopic intrinsic wave function of the system is an eigenstate of this operation with eigenvalue $+1$ (see also [Bo 76 b]).

1.5.2 The Axially Symmetric Case

The Hamiltonian $\hat{H}_{\text{coll}}$ (1.54) is still very general. We shall restrict ourselves to cases of very pronounced minima in the potential surface at axially symmetric deformations $\beta=\beta_0$ and $\gamma=0$. We expect rotations and small vibrations of the nuclear surface. Expanding $T_{\text{rot}}-\hat{I}_3^2/2\mathcal{I}_3$ (1.55) around the point $\beta=\beta_0$, $\gamma=0$, we obtain in zeroth order the Hamiltonian of an axially symmetric rotor with the moment of inertia $\mathcal{I}_0=\mathcal{I}_1(\beta_0,0)=\mathcal{I}_2(\beta_0,0)$,

$$T'_{\text{rot}}=\frac{\hat{\mathbf{I}}^2-\hat{I}_3^2}{2\mathcal{I}_0}. \tag{1.62}$$

First-order terms are proportional to the deviations $(\beta-\beta_0)$ and γ. They mix rotational and vibrational degrees of freedom (rotational–vibrational coupling terms) and will be neglected here. The only remaining term in H_{coll} that still couples rotations and vibrations is $\hat{I}_3^2/2\mathcal{I}_3$. It cannot be expanded, since $\mathcal{I}_3$ vanishes for $\gamma=0$. However (as T'_{rot}), it commutes with $\hat{I}_3$ and K is therefore a good quantum number.

We now have to distinguish

(i) K = 0 bands ($I_3=0$). In this case, the rotational and vibrational motions decouple. The wave function is of the type

$$|\Psi^I_{M0}\rangle=g_0(\beta,\gamma)|IM0\rangle. \tag{1.63}$$

They are eigenfunctions of the rotational part of the Hamiltonian (1.54). $\mathcal{R}_1$-symmetry (1.59) requires the spin sequence $I=0,2,4,\ldots$. A detailed investigation of the vibrational part of H_{coll} [EG 70, Vol. I, Chap. 6] shows that it is easier to handle in the variables a_{20} and a_{22} (1.13). In the first step one neglects terms in the potential $V(a_{20},a_{22})$, which couple these two degrees of freedom. In this case, the motion in the coordinate a_{20} (usually called β-vibration) decouples from the motion in the coordinate a_{22} (usually called γ-vibration). Axial symmetry with respect to the 3-axis is preserved by the β-vibration (quantum number n_β), but violated by the γ-vibration (quantum number n_γ). Both types of motion are shown qualitatively in Fig. 1.11.

Superimposed on each vibrational state (n_β,n_γ) is a rotational band. The spectrum is given by (see [EG 70, Vol. I, Chap. 6])

$$E_{n_\beta n_\gamma}(I)=E_{n_\beta n_\gamma}(0)+\frac{\hbar^2}{2\mathcal{I}_0}I\cdot(I+1) \tag{1.64}$$

with the band head

$$E_{n_\beta n_\gamma}(0)=\hbar\omega_\beta(n_\beta+1/2)+\hbar\omega_\gamma(2n_\gamma+1),$$
$$n_\beta=0,1,2,\ldots,\qquad n_\gamma=0,1,2,\ldots, \tag{1.65}$$

where ω_β and ω_γ are the frequencies of β- and γ-vibrations.

$$\omega_\beta=(C_{20}/B_2)^{1/2}\qquad \omega_\gamma=(C_{22}/B_2)^{1/2}$$

In fact, such bands have been observed in many even–even nuclei, in particular the ground state band ($n_\beta = n_\gamma = 0$) and the "β-band" ($n_\beta = 1$, $n_\gamma = 0$). However, as we have already discussed, the constants ϑ_0 and $\omega_\beta, \omega_\gamma$ of the hydrodynamical model do not agree with the experimental data.

(ii) $K \neq 0$ bands. Together with $\mathfrak{R}_1$-symmetry [Eq. (1.58)], we see that the wave function now has the form

$$|\Psi^I_{MK}\rangle = g_K(\beta, \gamma)\frac{1}{\sqrt{2}}\{|IMK\rangle + (-)^I|IM-K\rangle\}. \tag{1.66}$$

The $\mathfrak{R}_1$ and $\mathfrak{R}_2$ symmetries ((1.59) and (1.61)) give the selection rule: K must be even. Such $K \neq 0$ bands have, therefore, the spin sequence $I = |K|$, $|K|+1$, $|K|+2, \ldots$. The motion in a_{20} (β-vibration) can again be separated from the rest. However, the term $\hat{I}_3/2\vartheta_3 = K^2/16B_2a_{22}^2$ couples the γ-vibration with the rotation around the 3-axis. We can easily understand this fact if we realize that a γ-vibration can be represented as a superposition of two rotations of a triaxial nucleus around the 3-axis, but with opposite K-values [BM 75, p. 656].

The following spectrum is obtained [EG 70, Vol. I, Chap. 6]

$$E_{K,\, n_\beta,\, n_\gamma}(I) = E_{K,\, n_\beta,\, n_\gamma}(0) + \frac{\hbar^2}{2\vartheta_0}(I(I+1) - K^2) \tag{1.67}$$

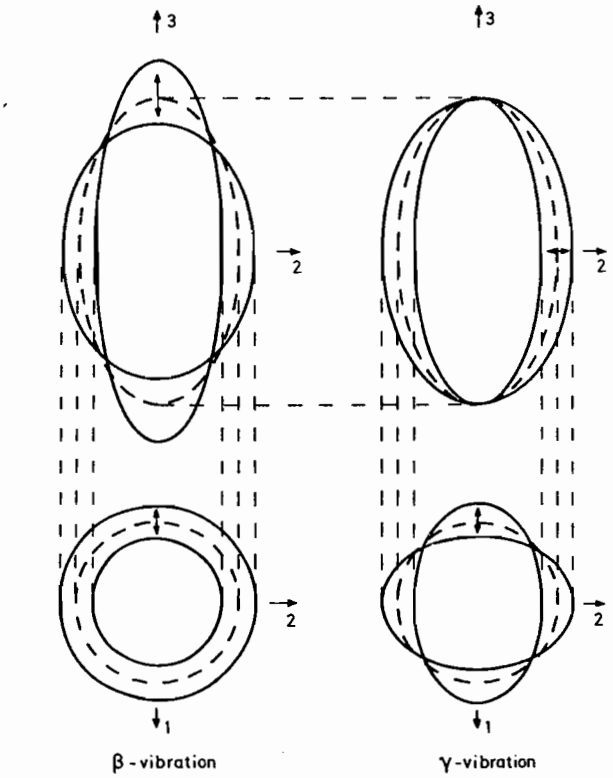


Figure 1.11. Schematical representation of β- and γ-vibrations by a cut along the (1, 3) and (1, 2) planes.

with the bandheads

$$E_{K,n_\beta,n_\gamma}(0)=\hbar\omega_\beta\left(n_\beta+\frac{1}{2}\right)+\hbar\omega_\gamma\left(2n_\gamma+1+\frac{|K|}{2}\right). \tag{1.68}$$

In fact, such bands have also been observed, especially the so-called "γ-band" in many deformed nuclei, which has the quantum numbers $K=2$, $n_\beta=0$, $n_\gamma=0$. This γ-band has the vibrational quantum number $n_\gamma=0$, however, one is not allowed to apply the classical picture of no vibration in this case. It would correspond to $\gamma=0$ and imply $\vartheta_3=0$, which would forbid a rotation with $K\neq 0$. Only the quantum mechanical zero point vibration in the γ-direction makes this possible.

Figure 1.12 shows the qualitative structure of the collective ($\lambda=2$) excitation in deformed and spherical nuclei.

The spherical nuclei have a harmonic spectrum. In the deformed nuclei we observe several rotational bands built on the ground state, on the β-vibrational state $K=0$, $n_\beta=1$, $n_\gamma=0$, and on the γ-vibrational state $K=2$, $n_\beta=n_\gamma=0$. However, these pure cases are not exactly realized in nature. In fact, we already observe in spherical nulcei a splitting of the two-boson triplet $(0^+,2^+,4^+)$ and in the deformed nuclei deviations from the $I(I+1)$ law. There is also a wide range of transitional nuclei in between these two limits. Going from isotope to isotope, one can sometimes observe a gradual transition from a vibrational to a rotational spectrum (for instance, in the Os region [SDG 76]). This is indicated by dashed lines in Fig. 1.12.

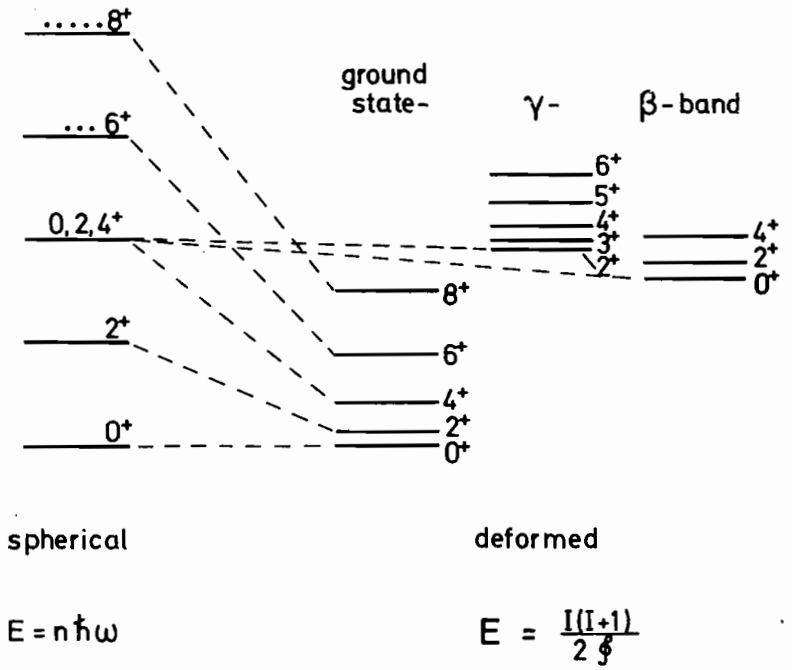


Figure 1.12. Schematic level schemes of spherical and deformed nuclei. (From [SDG 76].)

For a theoretical description of deviations from the $I\cdot(I+1)$ law, one has taken into account the rotational-vibrational coupling terms in $\hat{H}_{\text{coll}}$ which one gets by expanding ϑ_κ in powers of the deviations $(\beta-\beta_0)$ and γ[FG 62, FG 64, FGS 65, 66, ABP 68]. This rotation–vibration interaction causes changes in the moment of inertia of a band and corresponds to a change of the nuclear shape under the influence of the rotation (stretching effect). However, there also exist quite different excitations of nuclei in this energy region, for instance, two quasi-particle states and pairing vibrations (see Chap. 8), which have a much stronger influence on the rotational bands, and which are not taken into account in this simple model of a liquid drop with surface oscillations.

Before we leave this section, we want to discuss very briefly how to calculate electromagnetic moments and transition probabilities. In Eqs. (1.35) and (1.36) we defined the electric multipole and the magnetic moment operator in the coordinates $\alpha_{\lambda\mu}$. They are obviously written down in the laboratory system. In a deformed nucleus it is usually very easy to calculate the moments in the intrinsic system. To get the moments in the laboratory frame—the experimental values—one has to apply the transformation (1.11) of spherical tensors:

$$\hat{Q}^{\text{intr}}_{\lambda\mu}=\sum_{\mu'} D^{\lambda}_{\mu'\mu}\hat{Q}^{\text{lab}}_{\lambda\mu'}. \tag{1.69}$$

Since Q^{intr} does not depend on the Euler angles, we get from [Ed 57, Eqs. 4.6.2 and 5.4.1] the reduced matrix elements with respect to $|IMK\rangle$:

$$\langle I_1K_1\|\hat{Q}^{\text{lab}}_{\lambda}\|I_2K_2\rangle=\sum_{\mu'} Q^{\text{intr}}_{\lambda\mu'}(-)^{I_1-K_1}((2I_1+1)(2I_2+1))^{1/2}\begin{pmatrix} I_1 & \lambda & I_2\\ -K_1 & \mu' & K_2\end{pmatrix}. \tag{1.70}$$

We restrict ourselves now to pure K-bands and calculate only intraband $E2$-transitions ($n_{\gamma_1}=n_{\gamma_2}, n_{\beta_1}=n_{\beta_2}$). For the reduced matrix element we find

$$\langle n_\beta n_\gamma I_1K\|\hat{Q}^{\text{lab}}_2\|n_\beta n_\gamma I_2K\rangle = \sqrt{\frac{5}{16\pi}}\;Q_0(n_\beta,n_\gamma)\sqrt{(2I_1+1)(2I_2+1)}\;(-)^{I_1-K}\begin{pmatrix} I_1 & 2 & I_2\\ -K & 0 & K\end{pmatrix}, \tag{1.71}$$

where $Q_0(n_\beta,n_\gamma)$ is the intrinsic quadrupole moment of this band. In the ground state band with fixed β-value, and $\gamma=0$ we have [Eqs. (1.13) and (1.35)]

$$Q_0=\sqrt{\frac{16\pi}{5}}\;\frac{3}{4\pi}Ze\cdot R_0^2\cdot\beta. \tag{1.72}$$

From (B.73) and (1.71) we obtain, for example, for the so-called stretched $BE2$-values in a rotational band,

$$B(E2,I+2\to I)=Q_0^2\frac{5}{16\pi}\left|C^{I+2\;2\;I}_{K\;0\;K}\right|^2, \tag{1.73}$$

which, for $K=0$ bands, gives

$$B(E2, I+2\to I)=Q_0^2\frac{5}{16\pi}\frac{3}{2}\frac{(I+1)(I+2)}{(2I+3)(2I+5)}. \tag{1.74}$$

For the spectroscopic quadrupole moments $Q=\sqrt{16\pi/5}\,\langle IIK|\hat{Q}_{20}^{\text{lab}}|\,IIK\rangle$ [Eq. (B.32)], we get

$$Q=Q_0\frac{3K^2-I\cdot(I+1)}{(2I+3)(I+1)}. \tag{1.75}$$

The quotient of $Q:Q_0$ is the expectation value of $D_{00}^2=P_2(\cos\beta)$ in the state $M=I$. This means that one cannot measure the internal quadrupole moment Q_0, but only the value averaged over the rotational motion. In fact, Q_0 is not a physical observable. Its definition depends on the introduction of a body-fixed system which moves with the nucleus and has a model character. For the band head we usually have $I=K$. That means the spectroscopic quadrupole moment Q of ground states with $I=0$ vanishes and we can get information about Q_0 only for the excited states (for instance, by the reorientation effect in Coulomb excitation [BE 68], which gives the sign and the absolute value of Q_0). Another way to determine the absolute value of Q_0 is the measurement of the $B(E2)$-values (1.73) in the transitions within a band. Recent measurements up to high spin states [WCL 76, HJE 78] have shown that in many deformed nuclei the value of Q_0 stays fairly constant within these bands, even in cases where the spectrum shows deviations from the $I(I+1)$ character. This is a hint that these deviations are not caused by the change of deformation (stretching effect).

1.5.3 The Asymmetric Rotor

The rotational–vibrational interaction model discussed so far has been on the basis of a symmetric rotor. Further attempts to explain the deviations from the $I(I+1)$ law and the low-lying second 2^+ states in many nuclei have been undertaken by Davydov et al. using the picture of a pure triaxial rotor [DF 58, DR 59, Da 59, Da 65b]. As a first step they do not consider the vibrational excitations and diagonalize only the rotational energy (1.55). With the moments of inertia (1.48), this operator is proportional to β^{-2} and one can diagonalize it for all values of γ. The constant factor can afterwards be adjusted so as to reproduce the first 2^+ state. Using the symmetries $\mathfrak{R}_1$ and $\mathfrak{R}_2$ (1.59 and 1.61), the wave functions have the form

$$|\Psi_M^I\rangle=\sum_{K=0,\,2,\ldots,} g_K\{|IMK\rangle+(-)^I|IM-K\rangle\}. \tag{1.76}$$

Figure (1.13) shows the corresponding energy eigenvalues. For $\gamma=0°$ and $\gamma=60°$ one gets the $I(I+1)$ spectrum. Even for strong triaxial deformations one gets only slight deviations of this form. However, additional $2^+, 3^+, 4^+$ levels come down in energy. It is a characteristic of this structure to have a low-lying second 2^+ state. Although one can, with such a model, reproduce quite reasonably the experimental

data in some regions of the periodic table (for instance, for the Os-isotopes), strong objections can be raised to it:

(i) It is impossible to describe β-vibrations. In this case, one has to include additional vibrational degrees of freedom [DC 60]
(ii) Well pronounced minima in the energy surface that could justify a stable γ-deformation have not been found. Microscopic calculations for such transitional nuclei usually show only very shallow valleys in the γ direction [KB 68, ASS 69, GPA 72, GG 78].*

In recent years, stable γ-deformations have found new interest, because:

(i) Experimental data in odd–even transitional nuclei can be very well reproduced by a model of a particle coupled to an asymmetric rotor (see Chap. 5 and [MSD 74, Me 75, TF 75]).
(ii) Theoretical calculations for very high spin states (see Sec. 1.7 and [ALL 76]) show that nuclei can become triaxial in certain spin regions.

One can get a rough estimate of the level structure of triaxial nuclei by assuming a maximal triaxiality ($\gamma = 30°$). In this case, $\vartheta_2 = \vartheta_3 = \frac{1}{4}\vartheta_1 = \frac{1}{3}\vartheta_0$ (ϑ_0 is the moment of inertia at $\gamma = 0$), and we have symmetry about the 1-axis (in the kinetic energy). The projection α of I on to this axis is a good quantum number and one finds

$$E_\alpha(I) = \frac{3\hbar^2}{8\vartheta_0}(4I\cdot(I+1) - 3\alpha^2). \tag{1.77}$$

* Wilets and Jean [WJ 56] proposed a model which is completely γ-soft, i.e., $C_{22} = 0$ in Eq. (1.46).

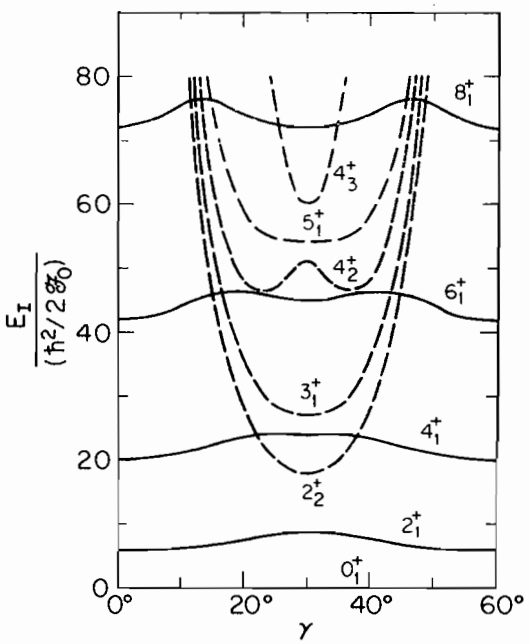


Figure 1.13. The energy eigenvalues of a deformed, asymmetric rotor with the hydrodynamic moments of inertia. (From [Me 75].)

$\mathfrak{R}_1$ and $\mathfrak{R}_2$ symmetry requires I and α to be even. For each I the level with the lowest energy (the so-called "yrast" level) has $\alpha = I$. Next in energy comes the band with $\alpha = I-2$, then $\alpha = I-4$, and so on. Therefore, we have a sequence of bands characterized by the "wobbling" quantum number $n = I-\alpha$ [BM 75]. One can calculate that the states with the same n are connected by large $E2$-transition probabilities. We get their spectrum from (1.77):

$$E_n(I) = \frac{3\hbar^2}{8\vartheta_0}(I(I+4)+3n(2I-n)). \tag{1.78}$$

However, one has to keep in mind that the structure of the spectrum (Fig. 1.13) depends drastically on the γ-dependence of the moments of inertia. There the hydrodynamical values have been used. For the rigid body values the spectrum would certainly look quite different.

1.6 Nuclear Fission

Up to now we have studied only small vibrations around the equilibrium shape in the liquid drop model. Shortly after the discovery of nuclear fission, attempts were made to understand this phenomenon using the concept of the nuclear drop [MF 39, Fr 39, BW 39].

In fact, a uniformly charged classical drop is only stable against fission (and spherical) if the Coulomb energy does not exceed a certain critical value. The Coulomb repulsion wants to deform the drop, the particles then being, on the average, further apart. The surface energy, on the contrary, being proportional to the surface of the drop, wants to keep it spherical. It is thus a subtle process of balance between these two effects (each being several hundred MeV in magnitude), which tells us whether there will be fission or not, according to a classical calculation. Of course, for such fission process, involving large deformations, one must go beyond the harmonic approximation discussed in the foregoing sections [BW 39].

The first step in describing the fission process [HW 53] is the choice of a suitable set of deformation parameters, which we call α. It has to be general enough to describe all the deformations that can occur. Cohen and Swiatecki [CS 62, 63], for instance, took as many as 18 multipolarities into account for the calculation of symmetric shapes of the form

$$R = R_0\left(1+\sum_{l=1}^{18}\alpha_l P_l(\cos\theta)\right). \tag{1.79}$$

This allows one to describe very general shapes. On the other hand, if one knows qualitatively the behavior of the droplet in the fission process, one is interested in introducing as few parameters as possible. Since one also wants to describe separated fragments, the set (1.79) is certainly not the most suitable one. For a realistic description of the fission process one needs at least three parameters [NS 65, Ni 72]:

(i) a parameter c, which describes in some way the length of the major semi-axis at the beginning of the fission process, and goes over into

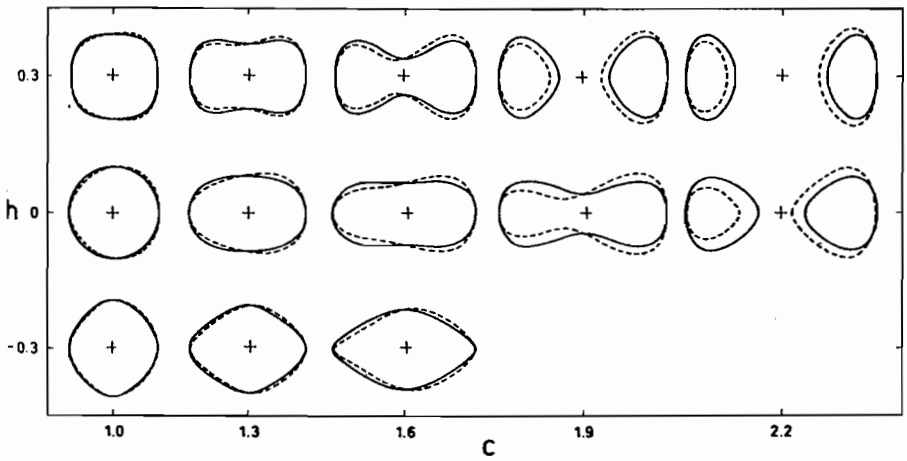


Figure 1.14. Some shapes in a suitable parameterization $\{c, h\}$. Solid lines show symmetric shapes. Dotted lines have a non-vanishing asymmetry parameter α_s. (Taken from [BDJ 72].)

the distance between the separated fragments (*elongation coordinate*);

(ii) a parameter h which characterizes the thickness of the neck between the fragments in formation (*neck coordinate*);

(iii) an asymmetry parameter α_s which measures the deviation from symmetry in the mass distribution of the separating fragments (in the case of asymmetric fission) (*fragmentation coordinate*).

Figure 1.14 shows an example for such a parameterization [BDJ 72].

In the next step, one has to calculate the *Hamiltonian* in the parameters α. The first part is the *potential energy* $V(\alpha)$. One usually measures it in units of the surface energy* of the corresponding sphere and gets

$$V(\alpha)=(B_S(\alpha)-1)+2x\cdot(B_C(\alpha)-1), \tag{1.80}$$

where B_S and B_C are the surface energy [in units of $E_S(0)$] and Coulomb energy [in units of $E_C(0)=\frac{3}{5}(eZ)^2/R_0$]. They are calculated as in Eqs. (1.32) and (1.33). However, now one can no longer suppose small deformations and must carry out the integrals explicitly. B_S and B_C depend only on the geometry. The only parameter that characterizes the nucleus is the so-called fissibility parameter x. We find

$$x=\frac{E_C(0)}{2E_S(0)}=\frac{Z^2/A}{(Z^2/A)_{\text{crit}}}, \tag{1.81}$$

*As we have discussed earlier, the potential energy can have a volume dependence corresponding to the intrinsic kinetic energy. For the large deformations implied in the fission process the dynamics can, however, have an important effect on $V(\alpha)$: Usually one assumes that the process of spontaneous fission is so slow that the nucleons have time to adjust their momentum distribution in such a way that it gives minimal total energy at each deformation. This momentum distribution is, of course, the spherical one. (For a more detailed discussion of this point, see Chap. 13.)

where

$$\left(\frac{Z^2}{A}\right)_{\text{crit}} = \frac{40\pi}{3}\frac{r_0^3\sigma}{e^2} \approx 50. \tag{1.82}$$

The calculations of Cohen and Swiatecki [CS 62, 63] show that the easiest way to deform the nucleus is the quadrupole deformation, and that the nuclear drop in general either stays spherical or fissions. Therefore, we can already decide, at zero deformation, whether or not fission occurs according to whether the curvature of the potential surface $V(\alpha)$ at the origin in the quadrupole direction is positive or negative. This, however, we have already calculated in considering the Bohr Hamiltonian [Eq. (1.17)] up to quadratic terms. Indeed, we see that the stiffness coefficient C_2 [Eq. (1.34)] is given by the difference of the surface and the Coulomb part of the deformation energy and C_2 starts to go negative for [Wi 64]

$$\frac{Z^2}{A} = \left(\frac{Z^2}{A}\right)_{\text{crit}} \quad \text{or} \quad x = 1. \tag{1.83}$$

This means that the classical droplet stays stable and spherical for $Z^2/A \lesssim 50$ or $x < 1$. For $x > 1$, it fissions immediately. This can be verified experimentally in giving increasing charge on a mercury drop.

For ^{238}U the mass formula (1.4) gives a Coulomb energy $E_C(0) = 830$ MeV and a surface energy $E_S(0) = 520$ MeV; for the fissibility parameter, we therefore obtain $x \simeq 0.8$, which is a typical value for the mass region

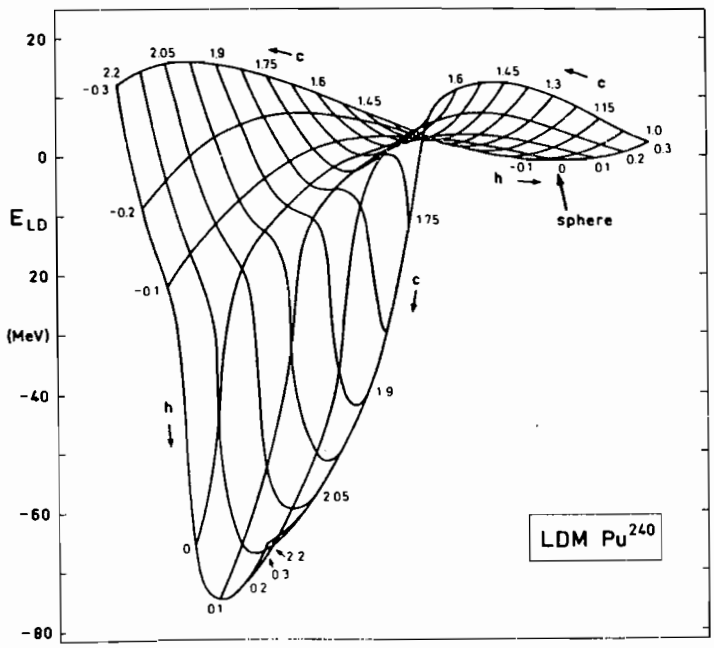


Figure 1.15. Perspective plot of the liquid drop model energy surface of ^{240}Pu. h describes the elongation and c the necking degree of freedom. (From [BDJ 72].)

$230 < A < 240$. Therefore, the classical liquid drop model cannot describe spontaneous fission. Observed spontaneous fission is thus due to the quantum mechanical tunnelling effect.

The potential energy surface (1.80) of the liquid drop has, in general, a complicated structure. For $x < 1$, we expect a minimum corresponding to a sphere, and for large distances outgoing valleys corresponding to different fragmentations [NS 65, Ni 69, MN 76].

Considering, for instance, only symmetric fission, we get for the two parameters c and h a surface as shown in Fig. 1.15. We see that it develops a well-defined saddle. Outside the saddle, the deformation energy falls off very steeply.

Cohen and Swiatecki [CS 63] have calculated the saddle-point shapes for many x-values with the ansatz (1.79). Some of them are shown in Fig. 1.16.

The meaning of these saddle-point shapes is that for a given x the drop would inevitably undergo fission if one, roughly speaking, elongates it a little bit more; conversely, it would fall back into its spherical shape if it was shortened a little bit. Going over the saddle along a line of steepest descent, the "bottom" of the fission valley defines a one-dimensional path. The length along this path is usually called the fission coordinate s. Figure 1.17 shows schematically the potential energy along this coordinate.

The fission barrier E_B is typically only a few, say, 5–7 MeV high. (This should be compared to several thousand MeV of total binding energy). Far out, the slope of the curve is due purely to Coulomb repulsion of the two droplets and thus falls off quite steeply for large Z.

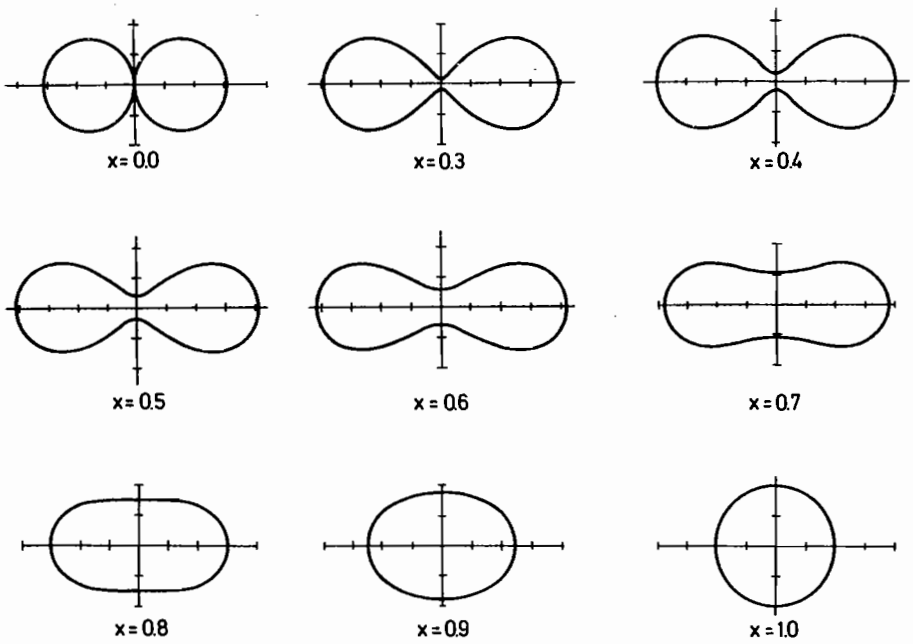


Figure 1.16. Saddle-point shapes for various values of the fissibility parameter x. (From [CS 63].)

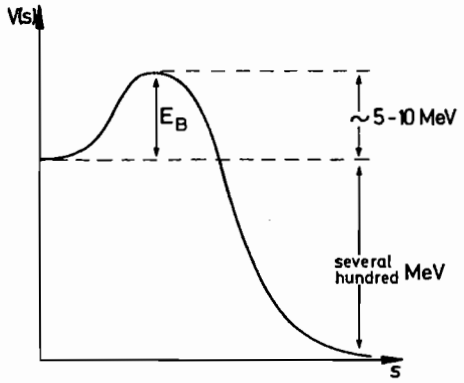


Figure 1.17. Schematic representation of the liquid drop fission barrier.

The next step in a dynamical description of fission would be the calculation of the *kinetic energy* (i.e., of mass parameters, which certainly depend on the deformation [HD 71, Gr 71]. Since the hydrodynamic model has not given good mass parameters for small oscillations, we cannot expect that it would work here either. We shall see in Section 12.3.7 how one usually calculates such quantities.

The final step would be a proper *quantization* and *solution* of the resulting Schrödinger equation. It goes far beyond the scope of this chapter to discuss such problems in detail. We only wish to mention that one usually calculates the lifetime for spontaneous fission from a WKB formula for a one-dimensional barrier penetration [Wi 64]. An essential ingredient of this formula is the shape of the potential $V(s)$, as shown in Fig. 1.17.

Unfortunately, the liquid drop model can in no way quantitatively describe fission, one of the major deficiencies being that the drop always fissions into two equally heavy fragments, whereas in fact nuclei quite often fission into unequal masses (asymmetric fission). From the fact that the fission barrier is a small number obtained from the difference of two huge numbers, we see that nuclear fission is an extremely subtle process, and we can easily imagine that, for example, quantum effects, like shell effects in the single particle motion (see Sec. 2.9), may very much affect it.

1.7 Stability of Rotating Liquid Drops

We have seen in Section 1.5 that among the vibrations of a liquid drop about a spherical surface there are rotation-like motions. In order to decouple them from the vibrations, however, we required stable finite deformations, which can only be explained by quantum mechanical shell effects. In fact, one finds rotational spectra in the experiment that justify such an assumption.

Heavy ion experiments allow the transfer of very large amounts of angular momentum (up to 80–100$\hbar$) to a nucleus. At such high excitations, one should expect that quantum effects no longer play any role, and one can then ask how a classical liquid drop having such angular velocities behaves.

In this section, we are therefore only interested in the purely classical problem of how a charged and incompressible liquid drop with a sharp surface and a certain surface energy behaves under rotation around a fixed axis in space. We are not interested in surface vibrations and therefore look for the stable shapes in this system.

Again we have to say something about the flow structure in this drop, and we shall assume rigid rotation; i.e., the moment of inertia is given by

$$\mathcal{I} = \mathcal{I}^{\text{rig}} = \int_V r_\perp^2 \rho_0 \, d^3r, \tag{1.84}$$

where $r_\perp$ is the distance of the volume d^3r from the rotational axis. As we shall see in Section 3.4, this assumption is very reasonable, because one expects the nuclear moment of inertia to approach the rigid value for high angular momenta.

The problem is very similar to the old astronomical problem of gravitating and incompressible homogeneous rotating masses. There the attractive force is the gravitation, which has the same structure (but the opposite sign) as the Coulomb force, and a surface force is neglected. Beginning with Newton, this problem has been studied by many famous mathematicians (for a historical review, see [Ch 69, Chap. 1]).

One of the results is that a gravitating droplet without surface tension has, for vanishing angular velocity ω, a spherical shape. With $\omega > 0$, it assumes first exact oblate spheroidal shapes where the rotational axis is an axis of symmetry (*Maclaurin shapes*). For increasing ω it flattens more and more, up to a certain angular velocity ω_I. For ω-values larger than ω_I, a new type of stable shape develops*—triaxial ellipsoids (*Jacobi shapes*)—and the drop rotates around the shortest principal axis. This point, ω_I, is called a bifurcation point, because for $\omega > \omega_I$ the Maclaurin shapes are still stationary, but they are no longer minima in all degrees of freedom [Ro 67]. One calls this *secularly unstable*. However, in uniformly rotating systems (gyrostatic systems) without friction forces, they can represent a stable motion (*ordinary stability*) [Ly 58]. Following the Jacobi shapes for higher ω-values, one reaches further bifurcation points. New shapes of stability develop and finally the droplet disintegrates.

Cohen, Plasil, and Swiatecki [CPS 74] investigated the problem of stability for the rotating nuclear liquid drop. They looked for the stationary surfaces of the energy

$$\begin{aligned} E(\alpha) &= E_S(\alpha) + E_C(\alpha) + E_R(\alpha) \\ &= E_S(0)(B_S(\alpha) + 2x \cdot B_C(\alpha) + yB_R(\alpha)). \end{aligned} \tag{1.85}$$

In addition to the surface (E_S) and the Coulomb (E_C) energies (see Eq. (1.32) f.), we now have the rotational energy at an angular momentum I:

$$E_R(\alpha) = \frac{\hbar^2 I^2}{2\mathcal{I}(\alpha)} = yE_S(0)B_R(\alpha). \tag{1.86}$$

The moment of inertia is given by Eq. (1.84): $\mathcal{I}(0) = \frac{2}{5}MR_0^2$. In addition to the fissibility parameter x, which measures the relation of Coulomb and surface energy, there is now the parameter

$$y = \frac{E_R(0)}{E_S(0)} = \frac{5}{16\pi} \frac{\hbar^2 I^2}{\sigma m r_0^4} \frac{1}{A^{7/3}} \approx 2I^2 A^{-7/3}, \tag{1.87}$$

*This corresponds to a second-order phase transition [BR 76].

which measures the relation of rotation and surface energy. The parameters x and y are determined by the properties of the nucleus and by the angular momentum one is interested in; again, the rest is pure geometry.

The result of these investigations [CPS 74] is that the nucleus behaves similarly to the gravitating droplet. At low angular momenta (low y-values), one gets flattened shapes with axial symmetry around the rotational axis. They are called *Hiskes shapes* and look similar to exact spheroids (Maclaurin shapes). They remain stable up to a critical value y_I, which corresponds to the *Jacobi point*. The value of y_I depends on x. For very heavy nuclei $x > 0.81$ the system fissions for $y > y_I$. For the rest of the nuclei a new kind of stable shape develops. These are called *Beringer–Knox shapes* [BK 61] and are similar to the Jacobi shapes. For very large angular momenta, these shapes get nearly axially symmetric around an axis perpendicular to the symmetry axis and finally become unstable against fission (y_{II}).

These calculations involve a large number of deformation parameters. One can describe the same features under the restriction of pure ellipsoidal shapes, whose half-axes a_1, a_2, a_3 are given by the two parameters β and γ^*:

$$a_\kappa = R_0 \exp\left\{ \sqrt{\frac{5}{4\pi}}\ \beta \cos\left(\gamma - \frac{2\pi}{3}\kappa\right)\right\} \qquad (\kappa = 1, 2, 3). \tag{1.88}$$

This definition guarantees volume conservation.

*We have used the Hill–Wheeler coordinates β and γ already for the description of small quadrupole deformations [Eq. (1.13)] of the nuclear surface. Here we define the nuclear surface as an exact ellipsoid by its half axis. This was the original definition of Hill and Wheeler [HW 53]. It agrees with Eq. (1.7) only for small β-values.

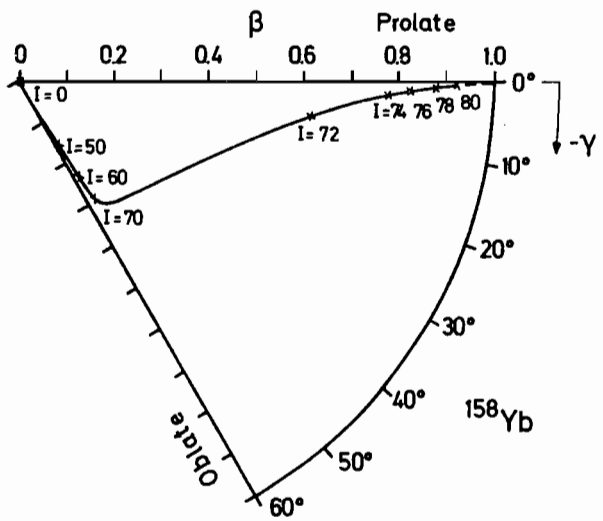


Figure 1.18. The equilibrium shapes (in terms of β and γ) of a rotating liquid drop as a function of angular momentum I. Since the 1-axis is the axis of rotation, we need negative γ-values, in contrast to the usual case where the three axes are equivalent. The nucleus ^{158}Yb has a value of $x = 0.618$. (From [ALL 76].)

It is relatively easy to calculate the rigid moment of inertia, and the surface and the Coulomb energies for these ellipsoids* as a function of β and γ and to look for the minimum of the energy surface for each value of I. Connecting these stable shapes, one gets a trajectory in the $\beta-\gamma$ plane, as in Fig. 1.18.

For each I, we have an energy surface whose local minima are investigated. As in the case of $I=0$ (Sec. 1.6), there also exists, besides the minimum, a fission valley with a saddle point. Because of the centrifugal force, the barrier height gets smaller for increasing angular momentum and a family of shapes becomes unstable against fission if the barrier disappears. As an example, we take the nucleus ^{127}La [CPS 74]. At angular momentum $I=0$ its LDM fission barrier is 40.0 MeV high. Its Jacobi point (y_J) corresponds to $I=67.8\hbar$. At this angular momentum it has an excitation energy of 44.31 MeV [$E_S(0)=49.5$ MeV] and the barrier height is only 7.8 MeV.

From this consideration it becomes clear that no nucleus can support more than a limiting angular momentum, otherwise it becomes unstable against fission. Figure 1.19 shows the angular momenta I_{II} at which the fission barrier vanishes as a function of the mass number A in the valley of beta stability [approximated by $N=A/2+0.2\cdot A^2/(200+A)$]. Light nuclei cannot support many units of angular momentum, because of their small size. Heavy nuclei have a reduced stability caused by the Coulomb energy. Nuclei with $A\simeq 130$ can reach the highest angular momenta of $\sim 100\hbar$. Experiments with heavy ions [BES 76] seem to be consistent with these limiting values of the liquid drop model.

* See, for instance, [Ca 61, CM 63].

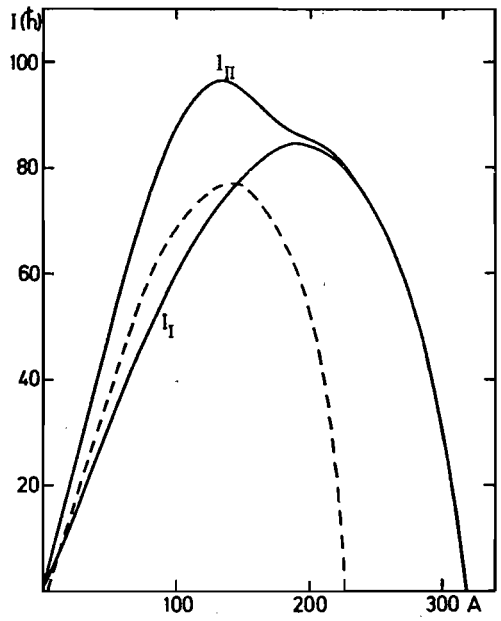


Figure 1.19. Maximal angular momentum I_{II} that a β-stable nucleus with mass number A can support, and the critical angular momentum I_I (Jacobi-bifurcation point). The dashed line corresponds to fission barrier heights of 8 MeV, which guarantees a reasonable lifetime against fission. (From [CPS 74].)